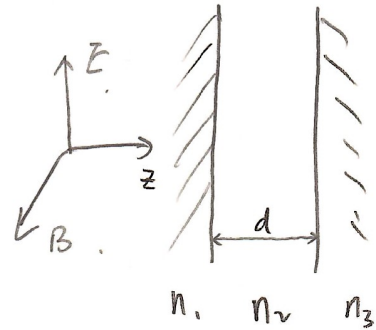


7.2 (a) The electromagnetic waves in the three regions are given by.

$$\text{I: } E_x(z, t) = E_i e^{i(k_1 z - \omega t)} + E_r e^{-i(k_1 z + \omega t)}$$

$$\text{II: } E_x(z, t) = E_1 e^{i(k_2 z - \omega t)} + E_2 e^{-i(k_2 z + \omega t)}$$

$$\text{III: } E_x(z, t) = E_t e^{i(k_3 z - \omega t)}$$



where $k_i = n_i \omega / c$. The boundary conditions at $z=0$ and $z=d$ dictate that

$$E_i + E_r = E_1 + E_2$$

$$n_1(E_i - E_r) = n_2(E_1 - E_2)$$

$$E_i e^{ik_1 d} + E_r e^{-ik_1 d} = E_t e^{ik_3 d}$$

$$n_2(E_i e^{ik_1 d} - E_r e^{-ik_1 d}) = n_3 E_t e^{ik_3 d}$$

The first two equations give

$$E_1 = \frac{1}{2} \left(1 + \frac{n_1}{n_2} \right) E_i + \frac{1}{2} \left(1 - \frac{n_1}{n_2} \right) E_r$$

$$E_2 = \frac{1}{2} \left(1 - \frac{n_1}{n_2} \right) E_i + \frac{1}{2} \left(1 + \frac{n_1}{n_2} \right) E_r$$

and last two equations give

$$E_1 = \frac{1}{2} \left(1 + \frac{n_3}{n_2} \right) E_t e^{i(k_3 - k_2)d}$$

$$E_2 = \frac{1}{2} \left(1 - \frac{n_3}{n_2} \right) E_t e^{i(k_3 + k_2)d}$$

Then, it can be shown that

$$E_t = \frac{2 e^{-ik_3 d}}{\left(1 + \frac{n_3}{n_1} \right) \cos(k_2 d) - i \left(\frac{n_2}{n_1} + \frac{n_3}{n_2} \right) \sin(k_2 d)} E_i$$

$$E_r = \frac{\left(1 - \frac{n_3}{n_1} \right) \cos(k_2 d) - i \left(\frac{n_2}{n_1} - \frac{n_3}{n_2} \right) \sin(k_2 d)}{\left(1 + \frac{n_3}{n_1} \right) \cos(k_2 d) - i \left(\frac{n_2}{n_1} + \frac{n_3}{n_2} \right) \sin(k_2 d)} E_i$$

Since reflected wave is in the same medium as the incident wave,

$$R = \left| \frac{E_r}{E_i} \right|^2 = \frac{\left(1 - \frac{n_3}{n_1} \right)^2 \cos^2(k_2 d) + \left(\frac{n_2}{n_1} - \frac{n_3}{n_2} \right)^2 \sin^2(k_2 d)}{\left(1 + \frac{n_3}{n_1} \right)^2 \cos^2(k_2 d) + \left(\frac{n_2}{n_1} + \frac{n_3}{n_2} \right)^2 \sin^2(k_2 d)}$$

The transmission coefficient is thus given by

$$T = 1 - R = \frac{4n_3/n_1}{\left(1 + \frac{n_3}{n_1}\right)^2 \cos^2(k_2 d) + \left(\frac{n_3}{n_2} + \frac{n_2}{n_1}\right)^2 \sin^2(k_2 d)}$$

(b) For $E_r = 0$, we need to have

$$\left(1 - \frac{n_3}{n_1}\right) \cos(k_2 d) = 0, \quad \left(\frac{n_3}{n_2} - \frac{n_1}{n_1}\right) \sin(k_2 d) = 0$$

Since n_1 and $n_3 = 1$ are fixed, we must have $\cos(k_2 d) = 0$ or equivalently

$$k_2 d = \frac{n_2 \omega_0 d}{c} = \frac{\pi}{2}, \quad d = \frac{c\pi}{2n_2 \omega_0}$$

For the second equation, $\sin(k_2 d) = 1$,

$$\frac{n_3}{n_2} - \frac{n_1}{n_1} = 0 \Rightarrow n_2 = \sqrt{n_1 n_3} = \sqrt{n_1}$$

and

$$d = \frac{c\pi}{2\omega_0 \sqrt{n_1}}$$